κChow-functions

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Abstract.

1. Introduction

1.1. Main results.

2. Framework

2.1. **Posets and Incidence Algebras.** Let $\mathcal{P}$ be a *finite partially ordered set* ranked by the function ρ. We denote by $\mathrm{I}(\mathcal{P})$ the incidence algebra of the poset $\mathcal{P}$ over the ring $\mathbb{Z}[x]$ with the algebra structure given by convolution. Some of the functions we consider in $\mathrm{I}(\mathcal{P})$ are δ; the zeta function, ζ; its inverse, the Möbius function, μ; the characteristic $\mathcal{P}$-kernel χ and its associated functions. Following [FMV24], we study some functions associated to χ. For example, the left and the right Kazhdan–Lusztig–Stanley functions (g and f respectively); the Z function; the reduced characteristic $\mathcal{P}$-kernel $\overline{\chi}$; the characteristic Chow function H; and the left and the right augmented Chow functions (G and F respectively). All these functions live inside a special subalgebra $\mathrm{I}_\rho(\mathcal{P})$ which consists of all the functions $\mathrm{a} \in \mathrm{I}(\mathcal{P})$ that satisfy $\deg(\mathrm{a}_{st}(x)) \leq \rho_{st} := \rho(t) - \rho(s)$ for all $s \leq t$ in $\mathcal{P}$. This subalgebra $\mathrm{I}_\rho(\mathcal{P})$ is equipped with an involution defined by $\mathrm{a} \mapsto \mathrm{a}^{\mathrm{rev}}$ with $\mathrm{a}^{\mathrm{rev}}_{st} := x^{\rho_{st}}\mathrm{a}_{st}(x^{-1})$. Special elements $\mathrm{a} \in \mathrm{I}_\rho(\mathcal{P})$ that satisfy $\mathrm{a}^{-1} = \mathrm{a}^{\mathrm{rev}}$ are known as $\mathcal{P}$-kernel. We summarize some properties of these functions as follows:

(1) $\chi = \mu \cdot \zeta^{\mathrm{rev}}$ is a $\mathcal{P}$-kernel,
(2) $g = \zeta^{\mathrm{rev}}$; $\zeta^{\mathrm{rev}}_{st} = x^{\rho_{st}}$,
(3) $\mathrm{Z} = g^{\mathrm{rev}} f = g f^{\mathrm{rev}}$ satisfies $\mathrm{Z}^{\mathrm{rev}} = \mathrm{Z}$,
(4) $\overline{\chi}_{st}(x) = \begin{cases} \chi_{st}(x)/(x-1) & \text{if } s < t, \\ -1 & \text{if } s = t, \end{cases}$
(5) $\mathrm{H} := -(\overline{\chi})^{-1}$ satisfies $\mathrm{H}_{st}(x) = x^{\rho_{st}-1}\mathrm{H}_{st}(x^{-1})$,
(6) $F = \mathrm{H} \cdot f^{\mathrm{rev}}$ satisfies $\mathrm{F}^{\mathrm{rev}} = \mathrm{F}$,
(7) $G = g^{\mathrm{rev}} \cdot \mathrm{H}$ satisfies $\mathrm{G}^{\mathrm{rev}} = \mathrm{G}$.

2.2. **The lorem ipsum modules.** Let $R := \mathbb{Z}[x, x^{-1}]$ be the ring of Laurent polynomials with integer coefficients. This ring contains the additive subgroups $\mathrm{Pal}(n)$, for each $n \in \mathbb{Z}$, consisting of all Laurent polynomials p satisfying $p(x) = x^n p(x^{-1})$. We note that these subgroups satisfy $\mathrm{Pal}(n) \cdot \mathrm{Pal}(m) \subset \mathrm{Pal}(n+m)$ and, in particular, $\mathrm{Pal}(0)$ is a ring.

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We now consider the module $\mathcal{H}(\mathcal{P})$ of $\mathcal{P}$ which is defined as the free R-module on the set $\{\mathrm{s}\}_{s\in\mathcal{P}}$. An element in $\mathcal{H}(\mathcal{P})$ is of the form

$$\alpha = \sum_{s\in\mathcal{P}} \alpha_s \cdot \mathrm{s}, \qquad \text{where } \alpha_s \in R.$$

We also define a family of R-linear map, $\{S_s : \mathcal{H}(\mathcal{P}) \to R\}_{s\in\mathcal{P}}$, given by

$$\sum_{t\in\mathcal{P}} \alpha_t \cdot \mathrm{t} \mapsto \sum_{t\geq s} x^{-\rho_{st}}\alpha_t.$$

Using these maps, we can now introduce some 'local' conditions to impose on the elements $\alpha \in \mathcal{H}(\mathcal{P})$:

(1) $\alpha_s \in \mathbb{Z}[x]$ for every $s \in \mathcal{P}$,
(2) $S_s(\alpha) \in \mathrm{Pal}(0)$ for every $s \in \mathcal{P}$.

Let $\mathcal{H}_p(\mathcal{P})$ the subgroup consisting of all elements $\alpha \in \mathcal{H}(\mathcal{P})$ that satisfy both of the conditions and let $\mathcal{H}'_p(\mathcal{P})$ be the subgroup consisting of only those that satisfy 2. We remark that the subgroup $\mathcal{H}_p(\mathcal{P})$ is a $\mathbb{Z}$-module and the subgroup $\mathcal{H}'_p(\mathcal{P})$ is a $\mathrm{Pal}(0)$-module. To further study these modules, we define two new R-bases of $\mathcal{H}(\mathcal{P})$ given by:

$$\mathrm{kl}^t = \sum_{s\leq t} \mathrm{kl}^t_s(x)\cdot \mathrm{s} := \sum_{s\leq t} x^{\rho_{st}} f_{st}(x^{-2})\cdot \mathrm{s}, \qquad \text{and} \qquad \mathrm{c}^t = \sum_{s\leq t} \mathrm{c}^t_s(x)\cdot \mathrm{s} := \sum_{s\leq t} x^{\rho_{st}} \mathrm{H}_{st}(x^{-2})\cdot \mathrm{s}.$$

We will name the first one as the **Kazhdan–Lusztig basis** and the second as the **Chow basis**. There are a few notes that we can make about these new bases:

(1) $\mathrm{kl}^t \in \mathcal{H}_p(\mathcal{P})$ for all $t \in \mathcal{P}$: We need to show that kl^t satisfies the two conditions. Firstly, since $\deg(f_{st}) < \rho_{st}/2$, we have $\mathrm{kl}^t_s \in \mathbb{Z}[x]$. Secondly, for a fixed $s \in \mathcal{P}$, we have

$$\begin{aligned} S_s(\mathrm{kl}^t) &= \sum_{u\geq s} x^{-\rho_{su}} \mathrm{kl}^t_u(x) \\ &= \sum_{u\geq s} x^{-\rho_{su}} x^{\rho_{ut}} f_{ut}(x^{-2}) \\ &= x^{\rho_{st}} \sum_{u:s\leq u\leq t} x^{-2\rho_{su}} f_{ut}(x^{-2}) \\ &= x^{\rho_{st}} \mathrm{Z}_{st}(x^{-2}). \end{aligned}$$

The factor $x^{\rho_{st}}$ is in $\mathrm{Pal}(2\rho_{st})$ and, since $\mathrm{Z}^{\mathrm{rev}} = \mathrm{Z}$, the factor $\mathrm{Z}_{st}(x^{-2})$ is in $\mathrm{Pal}(-2\rho_{st})$. Thus their product $S_s(\mathrm{kl}^t)$ is in $\mathrm{Pal}(2\rho_{st} - 2\rho_{st}) = \mathrm{Pal}(0)$, as required.

(2) $\mathrm{c}^t \in \mathcal{H}'_p(\mathcal{P})$: This requires $S_s(\mathrm{c}^t)$ to be in $\mathrm{Pal}(0)$ for every $s \in \mathcal{P}$. As in the previous case, $S_s(\mathrm{c}^t)$ now reduces to the product $x^{\rho_{st}} \left(g^{\mathrm{rev}} \cdot \mathcal{H}\right)_{st}(x^{-2}) = x^{\rho_{st}} G_{st}(x^{-2})$ which is also in $\mathrm{Pal}(0)$ since $G^{\mathrm{rev}} = G$.

Since the set $\{\mathrm{kl}^t\}_{t\in\mathcal{P}}$ (resp. $\{\mathrm{c}^t\}_{t\in\mathcal{P}}$) is R-independent, it is also a $\mathbb{Z}$ (resp. $\mathrm{Pal}(0)$)-independent set in $\mathcal{H}_p(\mathcal{P})$ (resp. $\mathcal{H}'_p(\mathcal{P})$). But we can have even more:

Theorem 2.1 *We have the following statements:*

(1) $\{\mathrm{kl}^t\}_{t\in\mathcal{P}}$ *forms a* $\mathbb{Z}$*-basis for* $\mathcal{H}_p(\mathcal{P})$,

(2) $\{\mathrm{c}^t\}_{t\in\mathcal{P}}$ *forms a* $\mathrm{Pal}(0)$*-basis for* $\mathcal{H}'_p(\mathcal{P})$.

Proof. We need to show that the sets span the respective modules.

Let $\alpha := \sum_{t\in\mathcal{P}} \alpha_t \cdot \mathrm{t}$ be an element in $\mathcal{H}_p(\mathcal{P})$. We write it as $\sum_{t\in\mathcal{P}} \beta_t \cdot \mathrm{kl}^t$ in the Kazhdan–Lusztig basis. Thus we have

$$\alpha = \sum_{t\in\mathcal{P}} \beta_t \cdot \mathrm{kl}^t = \sum_{t\in\mathcal{P}} \beta_t \cdot \sum_{s\le t} \mathrm{kl}^t_s \cdot \mathrm{s} = \sum_{s\in\mathcal{P}} \left(\sum_{t\ge s} \beta_t \mathrm{kl}^t_s\right) \cdot \mathrm{s} \implies \alpha_s = \sum_{t\ge s} \beta_t \mathrm{kl}^t_s. \tag{1}$$

We will show that $\beta_s \in \mathbb{Z}$ by induction on the corank of s.

Base case: When $s = \hat{1}$, $\alpha_{\hat{1}} = \beta_{\hat{1}} \mathrm{kl}^{\hat{1}}_{\hat{1}} = \beta_{\hat{1}}$.

Since $\alpha \in \mathcal{H}_p(\mathcal{P})$, we have that $S_{\hat{1}}(\alpha) = \sum_{t\ge\hat{1}} x^{-\rho_{\hat{1}t}} \alpha_t = \alpha_{\hat{1}} \in \mathrm{Pal}(0)$. But $\alpha_{\hat{1}}$ also needs to be in $\mathbb{Z}[x]$. Thus, $\beta_{\hat{1}} = \alpha_{\hat{1}} \in \mathrm{Pal}(0) \cap \mathbb{Z}[x] = \mathbb{Z}$ as required.

Inductive step: Assume that for all $t \in \mathcal{P}$ with $\rho(t) > \rho(s)$, we have $\beta_t \in \mathbb{Z}$. Then,

$$S_s(\alpha) = \sum_{t\ge s} x^{-\rho_{st}} \alpha_t.$$

Substituting for α_t using the change of basis formula in 1, we have

$$S_s(\alpha) = \sum_{t\ge s} x^{-\rho_{st}} \sum_{u\ge t} \beta_u \mathrm{kl}^u_t = \sum_{u\ge s} \beta_u \left(\sum_{t:s\le t\le u} x^{-\rho_{st}} \mathrm{kl}^u_t\right) = \sum_{u\ge s} \beta_u S_s(\mathrm{kl}^u).$$

Isolating the $u = s$ term, we have,

$$\beta_s = S_s(\alpha) - \sum_{u>s} \beta_u S_s(\mathrm{kl}^u).$$

The entire right-hand-side is in $\mathrm{Pal}(0)$ and thus $\beta_t \in \mathrm{Pal}(0)$.

Using the change of basis formula in 1, we also have $\beta_s = \alpha_s - \sum_{t>s} \beta_t \mathrm{kl}^t_s$. The right-hand-side here is just an expression of polynomials. Combining this fact with the previous, $\beta_s \in \mathrm{Pal}(0) \cap \mathbb{Z}[x] = \mathbb{Z}$.

Let us now show that $\{\mathrm{c}^t\}_{t\in\mathcal{P}}$ forms a $\mathrm{Pal}(0)$-basis for $\mathcal{H}'_p(\mathcal{P})$ using the same induction. Let $\alpha := \sum_{t\in\mathcal{P}} \alpha_t \cdot \mathrm{t}$ be an element in $\mathcal{H}'_p(\mathcal{P})$ which we write as $\sum_{t\in\mathcal{P}} \gamma_t \cdot \mathrm{c}^t$ in the Chow basis. We show that $\gamma_t \in \mathrm{Pal}(0)$ by induction on the corank of t.

When $t = \hat{1}$, we apply the map $S_{\hat{1}}$ to α:

$$S_{\hat{1}}(\alpha) = \sum_{u\in\mathcal{P}} \gamma_u S_{\hat{1}}(\mathrm{c}^u) = \gamma_{\hat{1}} \cdot 1 = \gamma_{\hat{1}}.$$

Since $\alpha \in \mathcal{H}'_p(\mathcal{P})$, we have $S_{\hat{1}}(\alpha) \in \mathrm{Pal}(0)$. Therefore, $\gamma_{\hat{1}} \in \mathrm{Pal}(0)$ proving the base case.

Now assume that for all $u \in \mathcal{P}$ with $\rho(u) > \rho(t)$, we have that $\gamma_u \in \mathrm{Pal}(0)$. We apply S_t to α:

$$S_t(\alpha) = \sum_{u\in\mathcal{P}} \gamma_u S_t(\mathrm{c}^u).$$

Since $S_t(\mathrm{c}^u)$ is non-zero only when $u \ge t$. We split the sum into the term $u = t$ and the terms $u > t$:

$$S_t(\alpha) = \gamma_t S_t(\mathrm{c}^t) + \sum_{u>t} \gamma_u S_t(\mathrm{c}^u).$$

We observe that $S_t(\mathrm{c}^t) = x^{-\rho_{tt}}\mathrm{c}_t^t = 1$. Solving for γ_t, we obtain:

$$\gamma_t = S_t(\alpha) - \sum_{u>t} \gamma_u S_t(\mathrm{c}^u).$$

On the right-hand side, we have that $S_t(\alpha) \in \mathrm{Pal}(0)$, $\gamma_u \in \mathrm{Pal}(0)$ by the inductive hypothesis, and $S_t(\mathrm{c}^u) \in \mathrm{Pal}(0)$ because $\mathrm{c}^u \in \mathcal{H}'_p(\mathcal{P})$, as established previously. Since $\mathrm{Pal}(0)$ is a ring, the linear combination of these terms remains in $\mathrm{Pal}(0)$. Thus, $\gamma_t \in \mathrm{Pal}(0)$. □

We continue our notes about the bases:

(3) $\mathrm{kl}_t^t = \mathrm{c}_t^t = 1 \in \mathbb{Z} \subset \mathrm{Pal}(0)$.

(4) $\mathrm{kl}_s^t \in x\mathbb{Z}[x]$ and $\mathrm{c}_s^t \in \mathrm{Pal}(2)$ for $s < t$ in $\mathcal{P}$: The first part is true because $\deg(f_{st}(x)) < \rho_{st}/2$. For the second assertion, using the fact that $\mathrm{H}_{st}(x)$ is a symmetric polynomial with center of symmetry $(\rho_{st} - 1)/2$, we note that

$$x^{\rho_{st}}\mathrm{H}_{st}(x^{-2}) \in \mathrm{Pal}(2\rho_{st}) \cdot \mathrm{Pal}(-2(\rho_{st} - 1)) \subset \mathrm{Pal}(2).$$

So we have $\mathrm{kl}^t \in \mathbb{Z} \cdot \mathrm{t} + \sum_{s<t} x\mathbb{Z}[x] \cdot \mathrm{s}$ and $\mathrm{c}^t \in \mathrm{Pal}(0) \cdot \mathrm{t} + \sum_{s<t} \mathrm{Pal}(2) \cdot \mathrm{s}$. These decompositions are quite special because the pairs $(\mathbb{Z}, x\mathbb{Z}[x])$ and $(\mathrm{Pal}(0), \mathrm{Pal}(2))$ form complementary subspaces of $\mathbb{Z}[x]$ and $\mathbb{Z}[x, x^{-1}]$ respectively. In other words,

$$\mathbb{Z}[x] = \mathbb{Z} \oplus x\mathbb{Z}[x] \qquad \text{and} \qquad \mathbb{Z}[x, x^{-1}] = \mathrm{Pal}(0) \oplus \mathrm{Pal}(2)$$

as $\mathbb{Z}$-modules and $\mathrm{Pal}(0)$-modules respectively. An element $f \in \mathbb{Z}[x, x^{-1}]$ can be uniquely written as

$$f = f_0 + f_2, \tag{2}$$

where $f_0 \in \mathrm{Pal}(0)$ and $f_2 \in \mathrm{Pal}(2)$. The projections are given by

$$f_0(x) = \frac{x^2 f(x^{-1}) - f(x)}{x^2 - 1} \quad \text{and} \quad f_2(x) = \frac{f(x) - f(x^{-1})}{1 - x^{-2}}.$$

Theorem 2.2 *If $\alpha \in \mathcal{H}'_p(\mathcal{P})$ is represented by $\alpha = \sum_{s\in\mathcal{P}} \alpha_s \cdot \mathrm{s}$ in the standard basis, then it is represented by $\alpha = \sum_{s\in\mathcal{P}} (\alpha_s)_0 \cdot \mathrm{c}^s$ in the Chow basis where $(\alpha_s)_0$ is the $\mathrm{Pal}(0)$ component of α_s from the decomposition in 2.*

Proof. Let $\alpha \in \mathcal{H}'_p(\mathcal{P})$ be represented by $\alpha = \sum_{s\in\mathcal{P}} \alpha_s \cdot \mathrm{s}$ in the standard basis and $\alpha = \sum_{s\in\mathcal{P}} \gamma_s \cdot \mathrm{c}^s$ in the Chow basis. We need to show that $\gamma_s = (\alpha_s)_0$. We prove this by induction on the corank of s.

For the base case, when $s = \hat{1}$, we have

$$S_{\hat{1}}(\alpha) = \alpha_{\hat{1}} \quad \text{and} \quad S_{\hat{1}}(\alpha) = \sum_{u\in\mathcal{P}} \gamma_u S_{\hat{1}}(\mathrm{c}^u) = \gamma_{\hat{1}} \cdot 1 = \gamma_{\hat{1}}.$$

Since $\alpha_{\hat{1}} \in \mathrm{Pal}(0)$, we have $\gamma_{\hat{1}} = \alpha_{\hat{1}} = (\alpha_{\hat{1}})_0$.

Now assume that for all $u \in \mathcal{P}$ with $\rho(u) > \rho(s)$, we have that $\gamma_u = (\alpha_u)_0$. Using the change of basis formula, we have

$$\alpha_s = \gamma_s \mathrm{c}_s^s + \sum_{u>s} \gamma_u \mathrm{c}_s^u.$$

Since $\mathrm{c}_s^u \in \mathrm{Pal}(2)$ for every $u > s$, we have $\sum_{u>s} \gamma_u \mathrm{c}_s^u \in \mathrm{Pal}(2)$ by using the inductive hypothesis. Since the remaining term $\gamma_s \mathrm{c}_s^s = \gamma_s$ is in $\mathrm{Pal}(0)$, it must be the unique $\mathrm{Pal}(0)$ component of α_s by the decomposition in 2, i.e., $\gamma_s = (\alpha_s)_0$. □

2.3. **Deletion of elements in the posets.** Now we will consider only those posets that are geometric lattices. These are the posets that are isomorphic to the lattice of flats of a matroid. We will denote the lattice of flats of a matroid M on a ground set E by $\mathcal{L}(\mathsf{M})$ and the rank function of M by rk_{M}. For a flat $F \in \mathcal{L}(\mathsf{M})$, we will denote the upper interval (the contraction) $[F, E]$ by M_F and the lower interval (the restriction)$[\emptyset, F]$ by M^F.

Let $e \in E$ be an element. We will denote the deletion of e by $\mathsf{M} \setminus e$. This is the matroid on the ground set $E \setminus \{e\}$ whose flats are the sets $F \setminus \{e\}$ for $F \in \mathcal{L}(\mathsf{M})$. There is a surjective map $\mathcal{L}(\mathsf{M}) \to \mathcal{L}(\mathsf{M} \setminus e)$ given by $F \mapsto F \setminus \{e\}$.

Definition 2.3 For a flat $F \in \mathcal{L}(\mathsf{M})$, we define the *discrepancy* $\delta(F)$ as:

$$\delta(F) := \mathrm{rk}_{\mathsf{M}}(F) - \mathrm{rk}_{\mathsf{M}\setminus e}(F \setminus e).$$

We define an operator $\Delta : \mathcal{H}(\mathsf{M}) \to \mathcal{H}(\mathsf{M} \setminus e)$ as the R-linear map given by:

$$\Delta(\mathrm{F}) := x^{-\delta(F)} \cdot \mathrm{F} \setminus \mathrm{e}.$$

Theorem 2.4 *Let $H \in \mathcal{L}(\mathsf{M} \setminus e)$ and denote the closure of H in M by $\overline{H}$. We have that $S_H^{\mathsf{M}\setminus e} \circ \Delta = S_{\overline{H}}^{\mathsf{M}}$. In particular, $\Delta(\mathrm{c}^F) \in \mathcal{H}'_p(\mathsf{M} \setminus e)$ for all $F \in \mathcal{L}(\mathsf{M})$.*

Proof. Let $\alpha \in \mathcal{H}(\mathsf{M})$ be represented by $\alpha = \sum_{F \in \mathcal{L}(\mathsf{M})} \alpha_F \cdot \mathrm{F}$ in the standard basis. We have that

$$S_H^{\mathsf{M}\setminus e}(\Delta(\alpha)) = \sum_{F \in \mathcal{L}(\mathsf{M})} \alpha_F S_H^{\mathsf{M}\setminus e}(x^{-\delta(F)} \cdot \mathrm{F} \setminus \mathrm{e}).$$

We can rewrite the sum as

$$S_H^{\mathsf{M}\setminus e}(\Delta(\alpha)) = \sum_{F \in \mathcal{L}(\mathsf{M})} \alpha_F S_H^{\mathsf{M}\setminus e}(x^{-\delta(F)} \cdot \mathrm{F} \setminus \mathrm{e})$$

$$S_H^{\mathsf{M}\setminus e}(\Delta(\alpha)) = \sum_{F \in \mathcal{L}(\mathsf{M})} \alpha_F S_H^{\mathsf{M}\setminus e}(x^{-\delta(F)} \cdot \mathrm{F} \setminus \mathrm{e})$$

□

References

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